



## **Semester I Examinations 2013/2014**

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|--------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Exam Codes</b>              | 3BA1, 3BS9, 3BS4, 4BS4, 3BPP2, 3MR3, 1OA1, 1EM1                                                                |
| <b>Exams</b>                   | Third Arts, Third Science, Fourth Science                                                                      |
| <b>Module</b>                  | Applied Statistics I                                                                                           |
| <b>Module Codes</b>            | <b>ST311</b>                                                                                                   |
| External Examiner              | Professor S. T. Buckland                                                                                       |
| Internal Examiner(s)           | Dr. E. Holian<br>Professor J. Hinde                                                                            |
| <b><u>Instructions:</u></b>    | <b>Answer any 3 questions from Questions 1-5</b><br><b>All Questions carry equal marks.</b>                    |
| <b>Duration</b>                | 2 Hours                                                                                                        |
| <b>No. of Pages</b>            | 22 pages, including this one                                                                                   |
| <b>School</b>                  | School of Mathematics, Statistics and Applied Mathematics                                                      |
| Release to Library:            | Yes                                                                                                            |
| <b><u>Requirements:</u></b>    |                                                                                                                |
| Statistical Tables/ Log Tables | Relevant distribution tables are attached to this paper.<br>The <i>New Formulae &amp; Tables</i> are optional. |
| Other Materials Allowed        | A calculator is allowed (non-programmable and not capable of storing text).                                    |

## Question 1

Complete parts (a), (b) and (c).

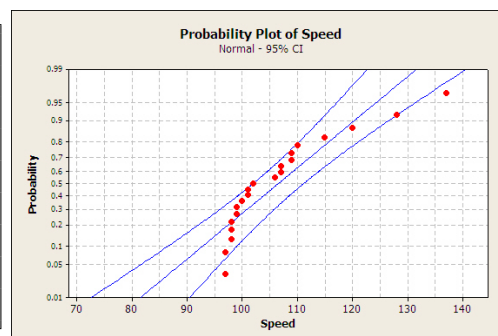
- (a) An analyst observed a sample of cars passing through a particular junction and recorded their speeds. The observations, recorded in kph, were:  
97, 97, 98, 98, 98, 99, 99, 100, 101, 101, 102, 106, 107, 107, 109, 109, 110, 115, 120, 128, 137.  
A summary of descriptive statistics and plots for this sample is provided below:

| Variable | N  | Mean   | SE Mean | StDev | Minimum | Q1    | Median | Q3     |
|----------|----|--------|---------|-------|---------|-------|--------|--------|
| Speed    | 21 | 106.57 | 2.34    | 10.74 | 97.00   | 98.50 | 102.00 | 109.50 |

| N for    |         |      |      |
|----------|---------|------|------|
| Variable | Maximum | Mode | Mode |
| Speed    | 137.00  | 98   | 3    |

| Letter Value Display: Speed |       |        |        |        |        |
|-----------------------------|-------|--------|--------|--------|--------|
|                             | Depth | Lower  | Upper  | Mid    | Spread |
| N=                          | 21    |        |        |        |        |
| M                           | 11.0  | 102.00 | 102.00 | 102.00 |        |
| H                           | 6.0   | 99.00  | 109.00 | 104.00 | 10.00  |
| E                           | 3.5   | 98.00  | 117.50 | 107.75 | 19.50  |
| D                           | 2.0   | 97.00  | 128.00 | 112.50 | 31.00  |
|                             | 1     | 97.00  | 137.00 | 117.00 | 40.00  |



- (i) By comparing the values of the mean, the median and the mode(s) for this sample what does this tell us about the shape of the distribution of the sample? [2 marks]
- (ii) Show how the letter values are calculated for this sample. [4 marks]
- (iii) For what purpose are the letter values used, and what do these particular letter values indicate? [4 marks]
- (iv) What does the normal probability plot suggest? [2 marks]
- (b) The advertised speed limit currently at this junction is 100kph. The analyst has been instructed to assess whether the typical speed for the population of all vehicles at this junction exceeds 100 kph.
- (i) It is suggested that a 1-sample t-test be carried out to infer whether the average speed for the population of all vehicles at this junction is greater than 100 kph. What can you conclude as a result of this analysis? Reference the output to justify your answer. [3 marks]

| One-Sample T: Speed       |    |        |       |         |                 |      |       |  |
|---------------------------|----|--------|-------|---------|-----------------|------|-------|--|
| Test of mu = 100 vs > 100 |    |        |       |         |                 |      |       |  |
| Variable                  | N  | Mean   | StDev | SE Mean | 95% Lower Bound | T    | P     |  |
| Speed                     | 21 | 106.57 | 10.74 | 2.34    | 102.53          | 2.81 | 0.005 |  |

Question 1 continues on the next page

Question 1 continued...

- (ii) Do you have any concerns about applying the 1-sample t-test in this application? If so, explain reasons for your concern. [2 marks]
- (iii) It is suggested that a 1-sample Signs test be carried out to infer whether the median speed for the population of all vehicles at this junction is greater than 100 kph.

|                                            |    |       |       |       |       |        |
|--------------------------------------------|----|-------|-------|-------|-------|--------|
| Sign Test for Median: Speed                |    |       |       |       |       |        |
| Sign test of median = 100.0 versus > 100.0 |    |       |       |       |       |        |
|                                            | N  | Below | Equal | Above | P     | Median |
| Speed                                      | 21 | 7     | 1     | 13    | 0.132 | 102.0  |

Verify, by calculation, the value of the p-value as printed in the output for the Signs test. Include explanation of the formulation of the p-value. [4 marks]

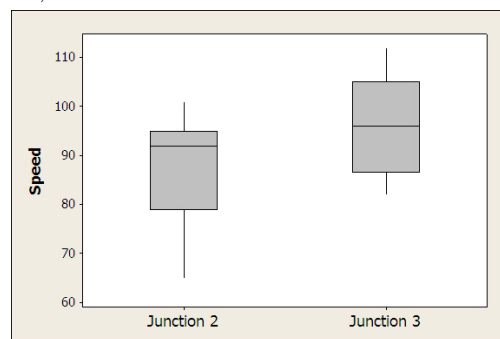
- (iv) If testing at  $\alpha = 0.05$ , what can you conclude as a result of the Signs test? Include a statement of the hypotheses being tested, your decision and conclusion in this application. [3 marks]

- (c) Two other junctions, Junction 2 and Junction 3, have the same speed limits. Junction 3, however, has proved to be an accident black-spot. Researchers want to assess whether the typical speed of vehicles approaching junction 2 is less than the typical speed of vehicles approaching junction 3. The speeds for a sample of 10 vehicles as they approached junction 2 were observed and the speeds for an independent sample of 9 vehicles as they approached junction 3 were observed. The observed speeds are listed below with some summary statistics and plots provided.

Junction 2 65, 73, 81, 89, 91, 93, 94, 95, 95, 101.

Junction 3 82, 84, 89, 90, 96, 100, 104, 106, 112.

|                         |    |       |       |        |
|-------------------------|----|-------|-------|--------|
| Descriptive Statistics: |    |       |       |        |
| Variable                | N  | Mean  | StDev | Median |
| Junction 2              | 10 | 87.70 | 11.26 | 92.00  |
| Junction 3              | 9  | 95.89 | 10.37 | 6.00   |



- (i) It is suggested to you to carry out a Wilcoxon Rank Sum test. Explain why, in this application, this test is more suitable than the 2-sample Z test, or the 2-sample t test. [4 marks]
- (ii) Carry out the test at significance level  $\alpha=0.05$ . Include a statement of the hypotheses being tested, formulation of the test statistic, the rejection region for the test statistic, and indicate clearly your decision and conclusion from the result of the test. [7 marks]

[Total 35 marks]

## Question 2

Complete parts (a), (b) and (c).

- (a) (i) "The true correlation coefficient between two variables  $X$  and  $Y$  for a population of individuals is known to have value  $\rho = 0$ . This implies that  $X$  and  $Y$  are unrelated."
- Is this statement true or false? If you deem the statement to be false give a reason for your answer. [2 marks]
- (ii) A sample of 25 individuals gave a sample correlation coefficient of  $r = 0.71$ . Carry out a hypothesis test of whether there is evidence of a significant increasing relationship between  $X$  and  $Y$  in the population. Clearly state the null and alternative hypotheses of your test, and give a test statistic, a rejection region, your decision and your conclusion. [6 marks]
- (b) Consider a simple linear regression model  $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$ .
- (i) What does the error term,  $\epsilon_i$ , represent? [2 marks]
- (ii) When carrying out an Analysis of Variance, explain in general what the terms  $SS_{Total}$ ,  $SS_{Reg}$  and  $SS_{Error}$  represent. [5 marks]
- (c) The following model is being considered in order to model the variability in a response variable  $Y$ , using five candidate predictor variables,  $X_1, X_2, X_3, X_4, X_5$ ,
- $$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + \beta_5 X_{5i} + \epsilon_i.$$
- The model is fitted to observations obtained for a sample of 25 individuals, and the results of the fit are provided in the partial printout on the next page. Use the output to answer the following questions.
- (i) Calculate the coefficient of determination  $R^2$  for this regression fit. Interpret the meaning of this value. [2 marks]
- (ii) Find the missing values in the ANOVA table, and complete the test, at the  $\alpha = 0.05$  level, including statements of the hypotheses being tested, a rejection region, a decision and conclusion. (*You do not have to calculate the p-value.*) [6 marks]
- (iii) Complete the missing t-test in the output, i.e. test whether  $X_2$  is a good predictor of  $Y$  at the  $\alpha = 0.05$  level. Include the hypotheses being tested, the calculation of the test statistic, a rejection region, a decision and conclusion. [5 marks]
- (iv) Use the partial F-test to test the hypothesis :  $H_0 : \beta_4 = \beta_5 = 0$  at  $\alpha = 0.05$ . Interpret your findings. [5 marks]
- (v) The output provides a prediction interval and a confidence interval for the response when  $X_1 = 7, X_2 = 10, X_3 = 1, X_4 = 10, X_5 = 28$ . Explain the difference as to when a prediction interval and a confidence interval applies. [2 marks]

Question 2 continues on the next page

Question 2 continued...

Regression Analysis: Y versus X1, X2, X3, X4, X5

The regression equation is

Y = - 5.53 + 2.96 X1 + 1.71 X2 - 9.15 X3 + 0.565 X4 + 0.717 X5

| Predictor | Coef  | SE Coef | T     | P     |
|-----------|-------|---------|-------|-------|
| Constant  | -5.53 | 22.68   | -0.24 | 0.810 |
| X1        | 2.96  | 1.870   | 1.58  | 0.130 |
| X2        | 1.71  | 0.3384  | -     | -     |
| X3        | -9.15 | 3.762   | -2.43 | 0.025 |
| X4        | 0.565 | 0.4417  | 1.28  | 0.216 |
| X5        | 0.717 | 0.4229  | 1.69  | 0.107 |

S = 10.0451    R-Sq = - %    R-Sq(adj) = - %

Analysis of Variance

| Source         | DF | SS     | MS | F | P |
|----------------|----|--------|----|---|---|
| Regression     | -  | 5247.5 | -  | - | - |
| Residual Error | -  | -      | -  |   |   |
| Total          | 24 | 7164.7 |    |   |   |

| Source | DF | Seq SS |
|--------|----|--------|
| X1     | 1  | 165.1  |
| X2     | 1  | 3701.6 |
| X3     | 1  | 706.2  |
| X4     | 1  | 384.9  |
| X5     | 1  | 289.7  |

Predicted Values for New Observations

New

| Obs | Fit   | SE Fit | 95% CI         | 95% PI         |
|-----|-------|--------|----------------|----------------|
| 1   | 60.15 | 2.34   | (55.26, 65.05) | (38.57, 81.74) |

Values of Predictors for New Observations

New

| Obs | X1   | X2   | X3   | X4   | X5   |
|-----|------|------|------|------|------|
| 1   | 7.00 | 10.0 | 1.00 | 30.0 | 28.0 |

[Total 35 marks]

### Question 3

Data obtained from a study exploring the relationship between graduation rates in colleges with a number of candidate predictor variables, was collected for a sample 45 colleges. The graduation rates, *GradRate*, were recorded as % of first year students that graduated within 5 years and the candidate predictor variables recorded were:

- *Accepted*, % of applicants accepted,
- *Alumnispons*, % of alumni giving sponsorship to the college,
- *StdFac*, student to faculty ratio,
- *Sat25*, the 25th percentile SAT score of first years entering the college,
- *Sat75*, the 75th percentile SAT score of first years entering the college,
- *Topten*, % of first years who were in the top 10% of their final year secondary school class.

- (a) A multiple linear regression model using three of the explanatory variables, *Accepted*, *Alumnispons* and *StdFac*, was first considered giving the following output:

The regression equation is  
 $\text{GradRate} = 83.7 - 0.252 \text{ Accepted} + 0.267 \text{ Alumnispons} + 0.218 \text{ StdFac}$

| Predictor   | Coef     | SE Coef | T     | P     |
|-------------|----------|---------|-------|-------|
| Constant    | 83.718   | 5.376   | 15.57 | 0.000 |
| Accepted    | -0.25192 | 0.04131 | -6.10 | 0.000 |
| Alumnispons | 0.26739  | 0.08265 | 3.24  | 0.002 |
| StdFac      | 0.2176   | 0.2192  | 0.99  | 0.327 |

S = 4.58009    R-Sq = 71.8%    R-Sq(adj) = 69.7%

#### Analysis of Variance

| Source         | DF | SS      | MS     | F     | P     |
|----------------|----|---------|--------|-------|-------|
| Regression     | 3  | 2191.13 | 730.38 | 34.82 | 0.000 |
| Residual Error | 41 | 860.07  | 20.98  |       |       |
| Total          | 44 | 3051.20 |        |       |       |

| Source      | DF | Seq SS  |
|-------------|----|---------|
| Accepted    | 1  | 1937.13 |
| Alumnispons | 1  | 233.33  |
| StdFac      | 1  | 20.67   |

- (i) Interpret the meaning of the estimated coefficients of the three predictor variables *Accepted*, *Alumnispons* and *StdFac*. [5 marks]
- (ii) Write down the hypotheses, in terms of the model coefficients,  $\beta$ 's, being tested in the ANOVA table for this model.

What do you conclude as a result of this test:

- in terms of the model coefficients,  $\beta$ 's, and
- in terms of the predictor variables?

[5 marks]

Question 3 continues on the next page

Question 3 continued...

- (iii) Comment on the significance of the three predictor variables *Accepted*, *Alumnispons* and *StdFac*. Use the output to support your answer. Include a statement of the hypotheses being tested for any hypothesis test you reference.

[6 marks]

- (b) A model with just the two predictor variables *Sat25* and *Sat75* is fitted to the data giving the following output:

```

The regression equation is
GradRate = 20.5 + 0.0821 Sat25 - 0.0260 Sat75

Predictor      Coef  SE Coef      T      P      VIF
Constant      20.50   20.40     1.01  0.321
Sat25          0.08211  0.04154    1.98  0.055  21.627
Sat75         -0.02604  0.04708   -0.55  0.583  21.627

S = 5.92400    R-Sq = 51.7%    R-Sq(adj) = 49.4%

Analysis of Variance

Source          DF      SS      MS      F      P
Regression       2  1577.26  788.63  22.47  0.000
Residual Error  42  1473.94   35.09
Total           44  3051.20

Source  DF   Seq SS
Sat25   1  1566.53
Sat75   1   10.74

```

Discuss the predictive power of the two variables, by giving your conclusions from the outputs of the t-tests and ANOVA test provided. Ensure you provide the hypotheses being tested for any test that you reference.

[8 marks]

- (c) The following output was produced using the best subsets routine in which five of the six predictors are considered.

- (i) Explain the meaning of the phrase “a parsimonious model”. [2 marks]  
(ii) Write down the most suitable model, i.e. the set of predictors, that you think best explains the variability in observed **GradRate**.

Justify your choice of model by explaining the role of  $s$ ,  $R^2$ , adjusted  $R^2$ , and Mallows’s  $C_p$  in choosing the best model.

[9 marks]

Question 3 continues on the next page

Question 3 continued...

| Response is GradRate |  |  |  |  |  |  |  |  |  |
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## Question 4

Complete parts (a), (b) and (c).

(a) When fitting a simple linear regression model to data what assumptions are required? [5 marks]

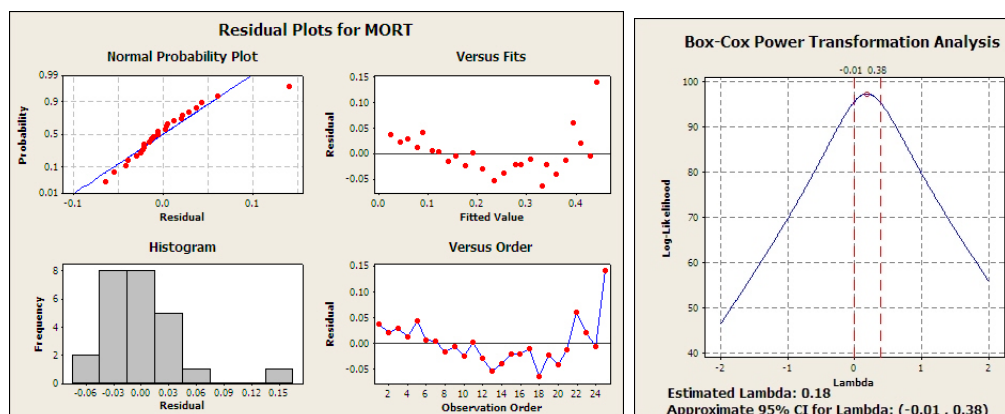
(b) In a study of the life-span of *Drosophila* fruitfly, being bred for increased life-span and resistance to stress, the mortality rate was observed every three or four days, by age measured in days, for a sample of 25 ages.

The output and diagnostic plots provided below show the results of a simple linear regression fit to the mortality rate, *Mortality*, using *Age* as a predictor.

The regression equation is  
 $Mortality = -0.126 + 0.00487 \text{ Age}$

| Predictor | Coef    | SE Coef | T     | P     |
|-----------|---------|---------|-------|-------|
| Constant  | -0.126  | 0.02564 | -4.91 | 0.000 |
| Age       | 0.00487 | 0.00032 | 14.92 | 0.000 |

S = 0.0430772    R-Sq = 90.6%    R-Sq(adj) = 90.2%



- Comment on the validity of the assumptions for this regression fit. [5 marks]
- What does the output of the Box-Cox transformation indicate? [5 marks]
- A simple linear regression using *Age* as the predictor was fitted to the transformed response variable  $\ln(Mortality)$  giving the output below:

The regression equation is  
 $\ln Mortality = -3.29 + 0.0226 \text{ Age}$

| Predictor | Coef   | SE Coef | T      | P     |
|-----------|--------|---------|--------|-------|
| Constant  | -3.29  | 0.06830 | -48.24 | 0.000 |
| Age       | 0.0226 | 0.00087 | 25.98  | 0.000 |

S = 0.114724    R-Sq = 96.7%    R-Sq(adj) = 96.6%

Question 4 continues on the next page

Question 4 continued...

Interpret the meaning of the estimated coefficient of *Age* in this model.

[5 marks]

- (iv) Using the model in part (iii) predict the mortality rate of a fly whose *Age* is observed to be 70 days.

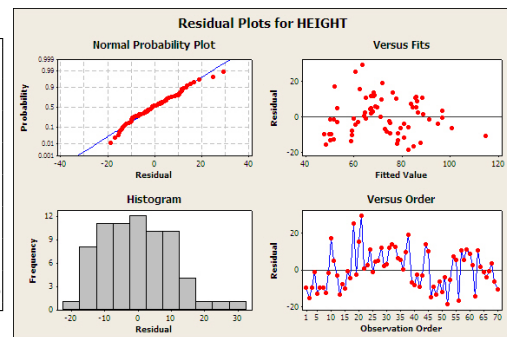
[4 marks]

- (c) In order to devise a method of estimating the height of Pine Trees, a model is being formulated to predict height based on the observed diameter of the tree measured at chest height. The diameter (inches) and actual height (feet) of a sample of 70 trees were observed and a simple linear regression fitted to the sample data giving the following output:

The regression equation is  
 $\text{Height} = 32.3 + 3.52 \text{ Diameter}$

| Predictor | Coef   | SE Coef | T     | P     |
|-----------|--------|---------|-------|-------|
| Constant  | 32.333 | 3.657   | 8.84  | 0.000 |
| Diameter  | 3.5224 | 0.3069  | 11.48 | 0.000 |

S = 10.5369    R-Sq = 65.9%    R-Sq(adj) = 65.4%



- (i) Comment on the validity of the assumptions using the diagnostic plots above.

[4 marks]

- (ii) A linear regression using *Diameter* and *DiameterSQ*, calculated as  $\text{Diameter}^2$ , as predictors was fitted to the response variable *Height* giving the following output:

The regression equation is  
 $\text{Height} = 15.0 + 6.85 \text{ Diameter} - 0.140 \text{ DiameterSQ}$

| Predictor  | Coef     | SE Coef | T     | P     |
|------------|----------|---------|-------|-------|
| Constant   | 15.011   | 7.746   | 1.94  | 0.057 |
| Diameter   | 6.852    | 1.359   | 5.04  | 0.000 |
| DiameterSQ | -0.14037 | 0.05590 | -2.51 | 0.014 |

S = 10.1485    R-Sq = 68.9%    R-Sq(adj) = 67.9%

Comment on the significance of the term *DiameterSQ*.

[4 marks]

- (iii) Using the model in part (ii) predict the height of a tree for which the diameter, *Diameter*, is 12 inches.

[3 marks]

[Total 35 marks]

## Question 5

Complete parts (a) and (b).

- (a) An agricultural engineering company has made two new types of lighting and housing for chickens. These make it possible to increase the amount of “daylight” which the birds receive with the aim of increasing the numbers of eggs laid. The treatments being compared are;

- E: extended day (14hours),
- F: flash lighting (natural day plus two 20-s flashes per night)
- and a control is also included, C: natural daylight.

The number of eggs laid by equal sized groups of birds observed over a number of weeks are recorded, for a sample of 18 farms, six farms set up under each of the three sets of conditions, C, E, F.

To investigate the effect of changes in conditions on egg numbers, indicator variables are defined for the three sets of conditions as:

$$\begin{aligned} IndC : & \begin{cases} 1 & \text{if number eggs were observed under control conditions C} \\ 0 & \text{otherwise} \end{cases} \\ IndE : & \begin{cases} 1 & \text{if number eggs were observed under conditions E} \\ 0 & \text{otherwise} \end{cases} \\ IndF : & \begin{cases} 1 & \text{if number eggs were observed under conditions F} \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

and the model,  $eggs_i = \beta_0 + \beta_1 IndC_i + \beta_2 IndE_i + \epsilon_i$ , is fitted to the data, giving:

|                                                  |       |         |       |       |
|--------------------------------------------------|-------|---------|-------|-------|
| The regression equation is                       |       |         |       |       |
| eggs = 339 - 37.8 IndC + 3.3 IndE                |       |         |       |       |
| Predictor                                        | Coef  | SE Coef | T     | P     |
| Constant                                         | 339   | 8.502   | 39.85 | 0.000 |
| IndC                                             | -37.8 | 12.02   | -3.15 | 0.007 |
| IndE                                             | 3.3   | 12.02   | 0.28  | 0.785 |
| S = 20.8257    R-Sq = 49.1%    R-Sq(adj) = 42.3% |       |         |       |       |

- i) Use the fitted model to predict the number of eggs laid for:
  - a farm that has control conditions *C*.
  - a farm that has treatment conditions *E*.
  - a farm that has treatment conditions *F*. [3 marks]
- ii) Interpret the meaning of the estimated coefficients of *IndC* and *IndE*, and comment on their significance. Ensure you state the hypotheses being tested when referring to the results of the output. [6 marks]
- iii) Write down an alternative model proposal that you would recommend be fitted to the observed data that would help assess the comparison of each of the treatment conditions, E and F, with the control conditions C. [2 marks]

Question 5 continues on the next page

Question 5 continued...

- (b) The bacterial count, measured as number of bacteria  $\times 10^6$  per ml, in cultures is believed to be linearly related to the age, in days, of the culture. Two culture types were tested to assess this phenomenon: aerated (extra air) and anaerobic (reduced air supply). The observed number of bacteria, the age and type of culture for a sample of 14 cultures were recorded.

- i) The regression analysis below models the response, *NumberBacteria*, by the predictor *Age*, and by an indicator variable *Culture* defined as

$$Culture = \begin{cases} 1 & \text{if the bacteria observed was grown in an aerobic culture} \\ 0 & \text{if the bacteria observed was grown in an anaerobic culture.} \end{cases}$$

The regression equation is  
Number Bacteria = 700 + 99.3 Age + 982 Culture

| Predictor | Coef  | SE Coef | T    | P     |
|-----------|-------|---------|------|-------|
| Constant  | 700.1 | 166.2   | 4.21 | 0.001 |
| Age       | 99.29 | 13.05   | 7.61 | 0.000 |
| Culture   | 981.7 | 138.6   | 7.08 | 0.000 |

S = 259.363    R-Sq = 90.8%    R-Sq(adj) = 89.1%

Interpret the meaning of the fitted coefficients of *Age* and *Culture* in this model. [4 marks]

- ii) Using the model fitted in part i), write down the estimated relationship between the number of bacteria and age for each of the two cultures. How do these differ? [3 marks]
- iii) An additional term was added to the model, which includes an interaction term, *CultureAge*, calculated as  $(Culture_i) \times (Age_i)$  for each individual, giving:

The regression equation is  
Number Bacteria = 1126 + 57.9 Age + 130 Culture + 82.8 CultureAge

| Predictor  | Coef    | SE Coef | T     | P     |
|------------|---------|---------|-------|-------|
| Constant   | 1125.78 | 65.77   | 17.12 | 0.000 |
| Age        | 57.910  | 5.682   | 10.19 | 0.000 |
| Culture    | 130.40  | 93.01   | 1.40  | 0.191 |
| CultureAge | 82.767  | 8.035   | 10.30 | 0.000 |

S = 79.8321    R-Sq = 99.2%    R-Sq(adj) = 99.0%

Analysis of Variance

| Source         | DF | SS      | MS      | F      | P     |
|----------------|----|---------|---------|--------|-------|
| Regression     | 3  | 7942397 | 2647466 | 415.41 | 0.000 |
| Residual Error | 10 | 63732   | 6373    |        |       |
| Total          | 13 | 8006129 |         |        |       |

Question 5 continues on the next page

*Question 5 continued...*

Write down the estimated relationship between the number of bacteria and age for each of the two cultures. How do these differ? [4 marks]

- iv) What hypotheses are being tested in the ANOVA table produced in this analysis? What can you conclude in terms of the predictors *Age*, *Culture* and *CultureAge*? [5 marks]
- v) Interpret the meaning of the coefficients of *Culture* and *CultureAge* and determine if these effects are significant, using the results of the individual t-tests in this analysis. Ensure you state the hypotheses being tested when referring to the results of the output.

What do the results tell you about the relationship between *NumberBacteria* and *Age* for the two different cultures? To illustrate your answer provide a rough sketch of any differences in the relationship between *NumberBacteria* and *Age* for the two different cultures. [8 marks]

[Total 35 marks]

## ST311 Formulae

Some useful z-scores:

$$Z_{0.1} = 1.28$$

$$Z_{0.05} = 1.64$$

$$Z_{0.025} = 1.96$$

$$Z_{0.01} = 2.33$$

$$Z_{0.005} = 2.58$$

Some ranges for the standard normal distribution:

$$1.35 \quad \text{middle } 50\% : Z_{0.75} - Z_{0.25}$$

$$2.30 \quad \text{middle } 75\% : Z_{0.875} - Z_{0.125}$$

$$3.07 \quad \text{middle } 87.5\% : Z_{0.9375} - Z_{0.0625}$$

$$3.73 \quad \text{middle } 93.75\% : Z_{0.96875} - Z_{0.03125}$$

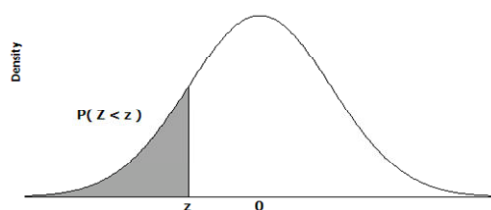
$$4.31 \quad \text{middle } 96.875\% : Z_{0.984375} - Z_{0.015625}$$

$$4.84 \quad \text{middle } 98.4375\% : Z_{0.9921875} - Z_{0.0078125}$$

Some formulae that may be useful when carrying out inference with parametric procedures:

- Inference for population correlation coefficient  $\rho$ :  $TS = r\sqrt{\frac{n-2}{1-r^2}}$
- Partial F-test :  $TS = \frac{(\text{additional SSreg})/q}{MS_{error}}$

Table: Cumulative Probabilities for the Standard Normal Distribution,  $Z \sim \text{Normal}(0,1)$ .



| <b>Z</b>    | <b>0.00</b> | <b>0.01</b> | <b>0.02</b> | <b>0.03</b> | <b>0.04</b> | <b>0.05</b> | <b>0.06</b> | <b>0.07</b> | <b>0.08</b> | <b>0.09</b> |
|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| <b>-3.4</b> | 0.0003      | 0.0003      | 0.0003      | 0.0003      | 0.0003      | 0.0003      | 0.0003      | 0.0003      | 0.0003      | 0.0002      |
| <b>-3.3</b> | 0.0005      | 0.0005      | 0.0005      | 0.0004      | 0.0004      | 0.0004      | 0.0004      | 0.0004      | 0.0004      | 0.0003      |
| <b>-3.2</b> | 0.0007      | 0.0007      | 0.0006      | 0.0006      | 0.0006      | 0.0006      | 0.0006      | 0.0005      | 0.0005      | 0.0005      |
| <b>-3.1</b> | 0.0010      | 0.0009      | 0.0009      | 0.0009      | 0.0008      | 0.0008      | 0.0008      | 0.0008      | 0.0007      | 0.0007      |
| <b>-3.0</b> | 0.0013      | 0.0013      | 0.0013      | 0.0012      | 0.0012      | 0.0011      | 0.0011      | 0.0011      | 0.0010      | 0.0010      |
| <b>-2.9</b> | 0.0019      | 0.0018      | 0.0018      | 0.0017      | 0.0016      | 0.0016      | 0.0015      | 0.0015      | 0.0014      | 0.0014      |
| <b>-2.8</b> | 0.0026      | 0.0025      | 0.0024      | 0.0023      | 0.0023      | 0.0022      | 0.0021      | 0.0021      | 0.0020      | 0.0019      |
| <b>-2.7</b> | 0.0035      | 0.0034      | 0.0033      | 0.0032      | 0.0031      | 0.0030      | 0.0029      | 0.0028      | 0.0027      | 0.0026      |
| <b>-2.6</b> | 0.0047      | 0.0045      | 0.0044      | 0.0043      | 0.0041      | 0.0040      | 0.0039      | 0.0038      | 0.0037      | 0.0036      |
| <b>-2.5</b> | 0.0062      | 0.0060      | 0.0059      | 0.0057      | 0.0055      | 0.0054      | 0.0052      | 0.0051      | 0.0049      | 0.0048      |
| <b>-2.4</b> | 0.0082      | 0.0080      | 0.0078      | 0.0075      | 0.0073      | 0.0071      | 0.0069      | 0.0068      | 0.0066      | 0.0064      |
| <b>-2.3</b> | 0.0107      | 0.0104      | 0.0102      | 0.0099      | 0.0096      | 0.0094      | 0.0091      | 0.0089      | 0.0087      | 0.0084      |
| <b>-2.2</b> | 0.0139      | 0.0136      | 0.0132      | 0.0129      | 0.0125      | 0.0122      | 0.0119      | 0.0116      | 0.0113      | 0.0110      |
| <b>-2.1</b> | 0.0179      | 0.0174      | 0.0170      | 0.0166      | 0.0162      | 0.0158      | 0.0154      | 0.0150      | 0.0146      | 0.0143      |
| <b>-2.0</b> | 0.0228      | 0.0222      | 0.0217      | 0.0212      | 0.0207      | 0.0202      | 0.0197      | 0.0192      | 0.0188      | 0.0183      |
| <b>-1.9</b> | 0.0287      | 0.0281      | 0.0274      | 0.0268      | 0.0262      | 0.0256      | 0.0250      | 0.0244      | 0.0239      | 0.0233      |
| <b>-1.8</b> | 0.0359      | 0.0351      | 0.0344      | 0.0336      | 0.0329      | 0.0322      | 0.0314      | 0.0307      | 0.0301      | 0.0294      |
| <b>-1.7</b> | 0.0446      | 0.0436      | 0.0427      | 0.0418      | 0.0409      | 0.0401      | 0.0392      | 0.0384      | 0.0375      | 0.0367      |
| <b>-1.6</b> | 0.0548      | 0.0537      | 0.0526      | 0.0516      | 0.0505      | 0.0495      | 0.0485      | 0.0475      | 0.0465      | 0.0455      |
| <b>-1.5</b> | 0.0668      | 0.0655      | 0.0643      | 0.0630      | 0.0618      | 0.0606      | 0.0594      | 0.0582      | 0.0571      | 0.0559      |
| <b>-1.4</b> | 0.0808      | 0.0793      | 0.0778      | 0.0764      | 0.0749      | 0.0735      | 0.0721      | 0.0708      | 0.0694      | 0.0681      |
| <b>-1.3</b> | 0.0968      | 0.0951      | 0.0934      | 0.0918      | 0.0901      | 0.0885      | 0.0869      | 0.0853      | 0.0838      | 0.0823      |
| <b>-1.2</b> | 0.1151      | 0.1131      | 0.1112      | 0.1093      | 0.1075      | 0.1056      | 0.1038      | 0.1020      | 0.1003      | 0.0985      |
| <b>-1.1</b> | 0.1357      | 0.1335      | 0.1314      | 0.1292      | 0.1271      | 0.1251      | 0.1230      | 0.1210      | 0.1190      | 0.1170      |
| <b>-1.0</b> | 0.1587      | 0.1562      | 0.1539      | 0.1515      | 0.1492      | 0.1469      | 0.1446      | 0.1423      | 0.1401      | 0.1379      |
| <b>-0.9</b> | 0.1841      | 0.1814      | 0.1788      | 0.1762      | 0.1736      | 0.1711      | 0.1685      | 0.1660      | 0.1635      | 0.1611      |
| <b>-0.8</b> | 0.2119      | 0.2090      | 0.2061      | 0.2033      | 0.2005      | 0.1977      | 0.1949      | 0.1922      | 0.1894      | 0.1867      |
| <b>-0.7</b> | 0.2420      | 0.2389      | 0.2358      | 0.2327      | 0.2296      | 0.2266      | 0.2236      | 0.2206      | 0.2177      | 0.2148      |
| <b>-0.6</b> | 0.2743      | 0.2709      | 0.2676      | 0.2643      | 0.2611      | 0.2578      | 0.2546      | 0.2514      | 0.2483      | 0.2451      |
| <b>-0.5</b> | 0.3085      | 0.3050      | 0.3015      | 0.2981      | 0.2946      | 0.2912      | 0.2877      | 0.2843      | 0.2810      | 0.2776      |
| <b>-0.4</b> | 0.3446      | 0.3409      | 0.3372      | 0.3336      | 0.3300      | 0.3264      | 0.3228      | 0.3192      | 0.3156      | 0.3121      |
| <b>-0.3</b> | 0.3821      | 0.3783      | 0.3745      | 0.3707      | 0.3669      | 0.3632      | 0.3594      | 0.3557      | 0.3520      | 0.3483      |
| <b>-0.2</b> | 0.4207      | 0.4168      | 0.4129      | 0.4090      | 0.4052      | 0.4013      | 0.3974      | 0.3936      | 0.3897      | 0.3859      |
| <b>-0.1</b> | 0.4602      | 0.4562      | 0.4522      | 0.4483      | 0.4443      | 0.4404      | 0.4364      | 0.4325      | 0.4286      | 0.4247      |
| <b>-0.0</b> | 0.5000      | 0.4960      | 0.4920      | 0.4880      | 0.4840      | 0.4801      | 0.4761      | 0.4721      | 0.4681      | 0.4641      |

**Table: Cumulative Probabilities for the Standard Normal Distribution,  $Z \sim \text{Normal}(0,1)$ .**

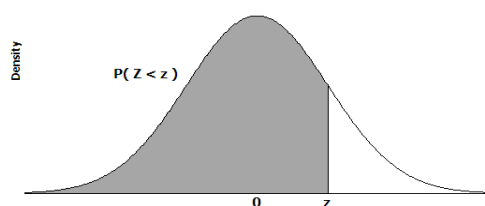
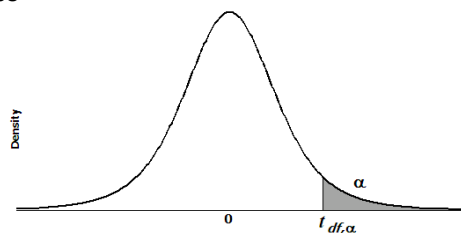
[illegible]



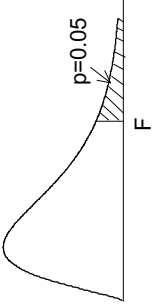
Table: t distribution critical values



| df   | Upper tail probability |       |       |       |       |        |        |        |        |         |         |         |
|------|------------------------|-------|-------|-------|-------|--------|--------|--------|--------|---------|---------|---------|
|      | 0.25                   | 0.20  | 0.15  | 0.10  | 0.05  | 0.025  | 0.02   | 0.01   | 0.005  | 0.0025  | 0.001   | 0.0005  |
| 1    | 1.000                  | 1.376 | 1.963 | 3.078 | 6.314 | 12.706 | 15.895 | 31.821 | 63.657 | 127.321 | 318.309 | 636.619 |
| 2    | 0.816                  | 1.061 | 1.386 | 1.886 | 2.920 | 4.303  | 4.849  | 6.965  | 9.925  | 14.089  | 22.327  | 31.599  |
| 3    | 0.765                  | 0.978 | 1.250 | 1.638 | 2.353 | 3.182  | 3.482  | 4.541  | 5.841  | 7.453   | 10.215  | 12.924  |
| 4    | 0.741                  | 0.941 | 1.190 | 1.533 | 2.132 | 2.776  | 2.999  | 3.747  | 4.604  | 5.598   | 7.173   | 8.610   |
| 5    | 0.727                  | 0.920 | 1.156 | 1.476 | 2.015 | 2.571  | 2.757  | 3.365  | 4.032  | 4.773   | 5.893   | 6.869   |
| 6    | 0.718                  | 0.906 | 1.134 | 1.440 | 1.943 | 2.447  | 2.612  | 3.143  | 3.707  | 4.317   | 5.208   | 5.959   |
| 7    | 0.711                  | 0.896 | 1.119 | 1.415 | 1.895 | 2.365  | 2.517  | 2.998  | 3.499  | 4.029   | 4.785   | 5.408   |
| 8    | 0.706                  | 0.889 | 1.108 | 1.397 | 1.860 | 2.306  | 2.449  | 2.896  | 3.355  | 3.833   | 4.501   | 5.041   |
| 9    | 0.703                  | 0.883 | 1.100 | 1.383 | 1.833 | 2.262  | 2.398  | 2.821  | 3.250  | 3.690   | 4.297   | 4.781   |
| 10   | 0.700                  | 0.879 | 1.093 | 1.372 | 1.812 | 2.228  | 2.359  | 2.764  | 3.169  | 3.581   | 4.144   | 4.587   |
| 11   | 0.697                  | 0.876 | 1.088 | 1.363 | 1.796 | 2.201  | 2.328  | 2.718  | 3.106  | 3.497   | 4.025   | 4.437   |
| 12   | 0.695                  | 0.873 | 1.083 | 1.356 | 1.782 | 2.179  | 2.303  | 2.681  | 3.055  | 3.428   | 3.930   | 4.318   |
| 13   | 0.694                  | 0.870 | 1.079 | 1.350 | 1.771 | 2.160  | 2.282  | 2.650  | 3.012  | 3.372   | 3.852   | 4.221   |
| 14   | 0.692                  | 0.868 | 1.076 | 1.345 | 1.761 | 2.145  | 2.264  | 2.624  | 2.977  | 3.326   | 3.787   | 4.140   |
| 15   | 0.691                  | 0.866 | 1.074 | 1.341 | 1.753 | 2.131  | 2.249  | 2.602  | 2.947  | 3.286   | 3.733   | 4.073   |
| 16   | 0.690                  | 0.865 | 1.071 | 1.337 | 1.746 | 2.120  | 2.235  | 2.583  | 2.921  | 3.252   | 3.686   | 4.015   |
| 17   | 0.689                  | 0.863 | 1.069 | 1.333 | 1.740 | 2.110  | 2.224  | 2.567  | 2.898  | 3.222   | 3.646   | 3.965   |
| 18   | 0.688                  | 0.862 | 1.067 | 1.330 | 1.734 | 2.101  | 2.214  | 2.552  | 2.878  | 3.197   | 3.610   | 3.922   |
| 19   | 0.688                  | 0.861 | 1.066 | 1.328 | 1.729 | 2.093  | 2.205  | 2.539  | 2.861  | 3.174   | 3.579   | 3.883   |
| 20   | 0.687                  | 0.860 | 1.064 | 1.325 | 1.725 | 2.086  | 2.197  | 2.528  | 2.845  | 3.153   | 3.552   | 3.850   |
| 21   | 0.686                  | 0.859 | 1.063 | 1.323 | 1.721 | 2.080  | 2.189  | 2.518  | 2.831  | 3.135   | 3.527   | 3.819   |
| 22   | 0.686                  | 0.858 | 1.061 | 1.321 | 1.717 | 2.074  | 2.183  | 2.508  | 2.819  | 3.119   | 3.505   | 3.792   |
| 23   | 0.685                  | 0.858 | 1.060 | 1.319 | 1.714 | 2.069  | 2.177  | 2.500  | 2.807  | 3.104   | 3.485   | 3.768   |
| 24   | 0.685                  | 0.857 | 1.059 | 1.318 | 1.711 | 2.064  | 2.172  | 2.492  | 2.797  | 3.091   | 3.467   | 3.745   |
| 25   | 0.684                  | 0.856 | 1.058 | 1.316 | 1.708 | 2.060  | 2.167  | 2.485  | 2.787  | 3.078   | 3.450   | 3.725   |
| 26   | 0.684                  | 0.856 | 1.058 | 1.315 | 1.706 | 2.056  | 2.162  | 2.479  | 2.779  | 3.067   | 3.435   | 3.707   |
| 27   | 0.684                  | 0.855 | 1.057 | 1.314 | 1.703 | 2.052  | 2.158  | 2.473  | 2.771  | 3.057   | 3.421   | 3.690   |
| 28   | 0.683                  | 0.855 | 1.056 | 1.313 | 1.701 | 2.048  | 2.154  | 2.467  | 2.763  | 3.047   | 3.408   | 3.674   |
| 29   | 0.683                  | 0.854 | 1.055 | 1.311 | 1.699 | 2.045  | 2.150  | 2.462  | 2.756  | 3.038   | 3.396   | 3.659   |
| 30   | 0.683                  | 0.854 | 1.055 | 1.310 | 1.697 | 2.042  | 2.147  | 2.457  | 2.750  | 3.030   | 3.385   | 3.646   |
| 40   | 0.681                  | 0.851 | 1.050 | 1.303 | 1.684 | 2.021  | 2.123  | 2.423  | 2.704  | 2.971   | 3.307   | 3.551   |
| 50   | 0.679                  | 0.849 | 1.047 | 1.299 | 1.676 | 2.009  | 2.109  | 2.403  | 2.678  | 2.937   | 3.261   | 3.496   |
| 60   | 0.679                  | 0.848 | 1.045 | 1.296 | 1.671 | 2.000  | 2.099  | 2.390  | 2.660  | 2.915   | 3.232   | 3.460   |
| 80   | 0.678                  | 0.846 | 1.043 | 1.292 | 1.664 | 1.990  | 2.088  | 2.374  | 2.639  | 2.887   | 3.195   | 3.416   |
| 100  | 0.677                  | 0.845 | 1.042 | 1.290 | 1.660 | 1.984  | 2.081  | 2.364  | 2.626  | 2.871   | 3.174   | 3.390   |
| 1000 | 0.675                  | 0.842 | 1.037 | 1.282 | 1.646 | 1.962  | 2.056  | 2.330  | 2.581  | 2.813   | 3.098   | 3.300   |
| Z*   | 0.674                  | 0.842 | 1.036 | 1.282 | 1.645 | 1.960  | 2.054  | 2.326  | 2.576  | 2.807   | 3.090   | 3.291   |

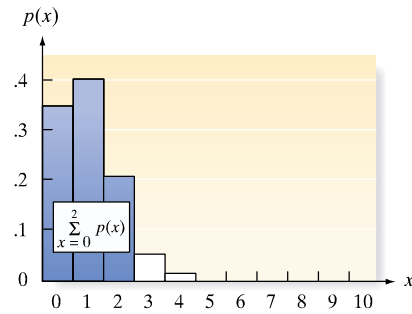
### F distribution for $p = 0.05$

For various pairs of degrees of freedom ( $v_1, v_2$ ), the tabled entries represent the critical values of F above which the proportion  $p = 0.05$  of the distribution falls. (Example, for  $df = 3, 8$  and  $F = 2.92$  is exceeded by  $p = 0.05$  of the population.



| $p = 0.05$ |         | Degrees of freedom for the numerator $v_1$ |         |         |         |         |         |         |         |         |         |         |         |         |         |         |         |          |  |  |
|------------|---------|--------------------------------------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|----------|--|--|
| $v_2$      | 1       | 2                                          | 3       | 4       | 5       | 6       | 7       | 8       | 9       | 10      | 12      | 15      | 20      | 30      | 40      | 60      | 120     | $\infty$ |  |  |
| 1          | 161.446 | 199.499                                    | 215.707 | 224.583 | 230.160 | 233.988 | 236.767 | 238.884 | 240.543 | 241.882 | 243.905 | 245.949 | 248.016 | 250.096 | 251.144 | 252.196 | 253.254 | 254.317  |  |  |
| 2          | 18.513  | 19.000                                     | 19.164  | 19.247  | 19.296  | 19.329  | 19.353  | 19.371  | 19.385  | 19.396  | 19.412  | 19.429  | 19.446  | 19.463  | 19.471  | 19.479  | 19.487  | 19.496   |  |  |
| 3          | 10.128  | 9.552                                      | 9.277   | 9.117   | 9.013   | 8.941   | 8.887   | 8.845   | 8.812   | 8.785   | 8.745   | 8.703   | 8.660   | 8.617   | 8.594   | 8.572   | 8.549   | 8.526    |  |  |
| 4          | 7.709   | 6.944                                      | 6.591   | 6.388   | 6.256   | 6.163   | 6.094   | 6.041   | 5.999   | 5.964   | 5.912   | 5.858   | 5.803   | 5.746   | 5.717   | 5.688   | 5.658   | 5.628    |  |  |
| 5          | 6.608   | 5.786                                      | 5.409   | 5.192   | 5.050   | 4.950   | 4.876   | 4.818   | 4.772   | 4.735   | 4.678   | 4.619   | 4.558   | 4.496   | 4.464   | 4.431   | 4.398   | 4.365    |  |  |
| 6          | 5.987   | 5.143                                      | 4.757   | 4.534   | 4.387   | 4.284   | 4.207   | 4.147   | 4.099   | 4.060   | 4.000   | 3.938   | 3.874   | 3.808   | 3.774   | 3.740   | 3.705   | 3.669    |  |  |
| 7          | 5.591   | 4.737                                      | 4.347   | 4.120   | 3.972   | 3.866   | 3.787   | 3.726   | 3.677   | 3.637   | 3.575   | 3.511   | 3.445   | 3.376   | 3.340   | 3.304   | 3.267   | 3.230    |  |  |
| 8          | 5.318   | 4.459                                      | 4.066   | 3.838   | 3.688   | 3.581   | 3.500   | 3.438   | 3.388   | 3.347   | 3.284   | 3.218   | 3.150   | 3.079   | 3.043   | 3.005   | 2.967   | 2.928    |  |  |
| 9          | 5.117   | 4.256                                      | 3.863   | 3.633   | 3.482   | 3.374   | 3.293   | 3.230   | 3.179   | 3.137   | 3.073   | 3.006   | 2.936   | 2.864   | 2.826   | 2.787   | 2.748   | 2.707    |  |  |
| 10         | 4.965   | 4.103                                      | 3.708   | 3.478   | 3.326   | 3.217   | 3.135   | 3.072   | 3.020   | 2.978   | 2.913   | 2.845   | 2.774   | 2.700   | 2.661   | 2.621   | 2.580   | 2.538    |  |  |
| 11         | 4.844   | 3.982                                      | 3.587   | 3.357   | 3.204   | 3.095   | 3.012   | 2.948   | 2.896   | 2.854   | 2.788   | 2.719   | 2.646   | 2.570   | 2.531   | 2.490   | 2.448   | 2.404    |  |  |
| 12         | 4.747   | 3.885                                      | 3.490   | 3.259   | 3.106   | 2.996   | 2.913   | 2.849   | 2.796   | 2.753   | 2.687   | 2.617   | 2.544   | 2.466   | 2.426   | 2.384   | 2.341   | 2.296    |  |  |
| 13         | 4.667   | 3.806                                      | 3.411   | 3.179   | 3.025   | 2.915   | 2.832   | 2.767   | 2.714   | 2.671   | 2.604   | 2.533   | 2.459   | 2.380   | 2.339   | 2.297   | 2.252   | 2.206    |  |  |
| 14         | 4.600   | 3.739                                      | 3.344   | 3.112   | 2.958   | 2.848   | 2.764   | 2.699   | 2.646   | 2.602   | 2.534   | 2.463   | 2.388   | 2.308   | 2.266   | 2.223   | 2.178   | 2.131    |  |  |
| 15         | 4.543   | 3.682                                      | 3.287   | 3.056   | 2.901   | 2.790   | 2.707   | 2.641   | 2.588   | 2.544   | 2.475   | 2.403   | 2.328   | 2.247   | 2.204   | 2.160   | 2.114   | 2.066    |  |  |
| 16         | 4.494   | 3.634                                      | 3.239   | 3.007   | 2.852   | 2.741   | 2.657   | 2.591   | 2.538   | 2.494   | 2.425   | 2.352   | 2.276   | 2.194   | 2.151   | 2.106   | 2.059   | 2.010    |  |  |
| 17         | 4.451   | 3.592                                      | 3.197   | 2.965   | 2.810   | 2.699   | 2.614   | 2.548   | 2.494   | 2.450   | 2.381   | 2.308   | 2.230   | 2.148   | 2.104   | 2.058   | 2.011   | 1.960    |  |  |
| 18         | 4.414   | 3.555                                      | 3.160   | 2.928   | 2.773   | 2.661   | 2.577   | 2.510   | 2.456   | 2.412   | 2.342   | 2.269   | 2.191   | 2.107   | 2.063   | 2.017   | 1.968   | 1.917    |  |  |
| 19         | 4.381   | 3.522                                      | 3.127   | 2.895   | 2.740   | 2.628   | 2.544   | 2.477   | 2.423   | 2.378   | 2.308   | 2.234   | 2.155   | 2.071   | 2.026   | 1.980   | 1.930   | 1.878    |  |  |
| 20         | 4.351   | 3.493                                      | 3.098   | 2.866   | 2.711   | 2.599   | 2.514   | 2.447   | 2.393   | 2.348   | 2.278   | 2.203   | 2.124   | 2.039   | 1.994   | 1.946   | 1.896   | 1.843    |  |  |
| 21         | 4.325   | 3.467                                      | 3.072   | 2.840   | 2.685   | 2.573   | 2.488   | 2.420   | 2.366   | 2.321   | 2.250   | 2.176   | 2.096   | 2.010   | 1.965   | 1.916   | 1.866   | 1.812    |  |  |
| 22         | 4.301   | 3.443                                      | 3.049   | 2.817   | 2.661   | 2.549   | 2.464   | 2.397   | 2.342   | 2.297   | 2.226   | 2.151   | 2.071   | 1.984   | 1.938   | 1.889   | 1.838   | 1.783    |  |  |
| 23         | 4.279   | 3.422                                      | 3.028   | 2.796   | 2.640   | 2.528   | 2.442   | 2.375   | 2.320   | 2.275   | 2.204   | 2.128   | 2.048   | 1.961   | 1.914   | 1.865   | 1.813   | 1.757    |  |  |
| 24         | 4.260   | 3.403                                      | 3.009   | 2.776   | 2.621   | 2.508   | 2.423   | 2.355   | 2.300   | 2.255   | 2.183   | 2.108   | 2.027   | 1.939   | 1.892   | 1.842   | 1.790   | 1.733    |  |  |
| 30         | 4.171   | 3.316                                      | 2.922   | 2.690   | 2.534   | 2.421   | 2.334   | 2.266   | 2.211   | 2.165   | 2.092   | 2.015   | 1.932   | 1.841   | 1.792   | 1.740   | 1.683   | 1.622    |  |  |
| 40         | 4.085   | 3.232                                      | 2.839   | 2.606   | 2.449   | 2.336   | 2.249   | 2.180   | 2.124   | 2.077   | 2.003   | 1.924   | 1.839   | 1.744   | 1.693   | 1.637   | 1.577   | 1.509    |  |  |
| 60         | 4.001   | 3.150                                      | 2.758   | 2.525   | 2.368   | 2.254   | 2.167   | 2.097   | 2.040   | 1.993   | 1.917   | 1.836   | 1.748   | 1.649   | 1.594   | 1.534   | 1.467   | 1.399    |  |  |
| 120        | 3.920   | 3.072                                      | 2.680   | 2.447   | 2.290   | 2.175   | 2.087   | 2.016   | 1.959   | 1.910   | 1.834   | 1.750   | 1.659   | 1.554   | 1.495   | 1.429   | 1.352   | 1.254    |  |  |
| $\infty$   | 3.841   | 2.996                                      | 2.605   | 2.372   | 2.214   | 2.099   | 2.010   | 1.938   | 1.880   | 1.831   | 1.752   | 1.666   | 1.571   | 1.459   | 1.394   | 1.318   | 1.221   | 1.000    |  |  |

TABLE II Binomial Probabilities



Tabulated values are  $\sum_{x=0}^k p(x)$ . (Computations are rounded at the third decimal place.)

a.  $n = 5$

| $\begin{matrix} p \\ k \end{matrix}$ | .01   | .05   | .10   | .20   | .30  | .40  | .50  | .60  | .70  | .80  | .90  | .95  | .99  |
|--------------------------------------|-------|-------|-------|-------|------|------|------|------|------|------|------|------|------|
| 0                                    | .951  | .774  | .590  | .328  | .168 | .078 | .031 | .010 | .002 | .000 | .000 | .000 | .000 |
| 1                                    | .999  | .977  | .919  | .737  | .528 | .337 | .188 | .087 | .031 | .007 | .000 | .000 | .000 |
| 2                                    | 1.000 | .999  | .991  | .942  | .837 | .683 | .500 | .317 | .163 | .058 | .009 | .001 | .000 |
| 3                                    | 1.000 | 1.000 | 1.000 | .993  | .969 | .913 | .812 | .663 | .472 | .263 | .081 | .023 | .001 |
| 4                                    | 1.000 | 1.000 | 1.000 | 1.000 | .998 | .990 | .969 | .922 | .832 | .672 | .410 | .226 | .049 |

b.  $n = 6$

| $\begin{matrix} p \\ k \end{matrix}$ | .01   | .05   | .10   | .20   | .30  | .40  | .50  | .60  | .70  | .80  | .90  | .95  | .99  |
|--------------------------------------|-------|-------|-------|-------|------|------|------|------|------|------|------|------|------|
| 0                                    | .941  | .735  | .531  | .262  | .118 | .047 | .016 | .004 | .001 | .000 | .000 | .000 | .000 |
| 1                                    | .999  | .967  | .886  | .655  | .420 | .233 | .109 | .041 | .011 | .002 | .000 | .000 | .000 |
| 2                                    | 1.000 | .998  | .984  | .901  | .744 | .544 | .344 | .179 | .070 | .017 | .001 | .000 | .000 |
| 3                                    | 1.000 | 1.000 | .999  | .983  | .930 | .821 | .656 | .456 | .256 | .099 | .016 | .002 | .000 |
| 4                                    | 1.000 | 1.000 | 1.000 | .998  | .989 | .959 | .891 | .767 | .580 | .345 | .114 | .033 | .001 |
| 5                                    | 1.000 | 1.000 | 1.000 | 1.000 | .999 | .996 | .984 | .953 | .882 | .738 | .469 | .265 | .059 |

c.  $n = 7$

| $\begin{matrix} p \\ k \end{matrix}$ | .01   | .05   | .10   | .20   | .30   | .40  | .50  | .60  | .70  | .80  | .90  | .95  | .99  |
|--------------------------------------|-------|-------|-------|-------|-------|------|------|------|------|------|------|------|------|
| 0                                    | .932  | .698  | .478  | .210  | .082  | .028 | .008 | .002 | .000 | .000 | .000 | .000 | .000 |
| 1                                    | .998  | .956  | .850  | .577  | .329  | .159 | .063 | .019 | .004 | .000 | .000 | .000 | .000 |
| 2                                    | 1.000 | .996  | .974  | .852  | .647  | .420 | .227 | .096 | .029 | .005 | .000 | .000 | .000 |
| 3                                    | 1.000 | 1.000 | .997  | .967  | .874  | .710 | .500 | .290 | .126 | .033 | .003 | .000 | .000 |
| 4                                    | 1.000 | 1.000 | 1.000 | .995  | .971  | .904 | .773 | .580 | .353 | .148 | .026 | .004 | .000 |
| 5                                    | 1.000 | 1.000 | 1.000 | 1.000 | .996  | .981 | .937 | .841 | .671 | .423 | .150 | .044 | .002 |
| 6                                    | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .998 | .992 | .972 | .918 | .790 | .522 | .302 | .068 |

(continued)

TABLE II (continued)

d.  $n = 8$ 

| $\begin{smallmatrix} p \\ k \end{smallmatrix}$ | .01   | .05   | .10   | .20   | .30   | .40  | .50  | .60  | .70  | .80  | .90  | .95  | .99  |
|------------------------------------------------|-------|-------|-------|-------|-------|------|------|------|------|------|------|------|------|
| 0                                              | .923  | .663  | .430  | .168  | .058  | .017 | .004 | .001 | .000 | .000 | .000 | .000 | .000 |
| 1                                              | .997  | .943  | .813  | .503  | .255  | .106 | .035 | .009 | .001 | .000 | .000 | .000 | .000 |
| 2                                              | 1.000 | .994  | .962  | .797  | .552  | .315 | .145 | .050 | .011 | .001 | .000 | .000 | .000 |
| 3                                              | 1.000 | 1.000 | .995  | .944  | .806  | .594 | .363 | .174 | .058 | .010 | .000 | .000 | .000 |
| 4                                              | 1.000 | 1.000 | 1.000 | .990  | .942  | .826 | .637 | .406 | .194 | .056 | .005 | .000 | .000 |
| 5                                              | 1.000 | 1.000 | 1.000 | .999  | .989  | .950 | .855 | .685 | .448 | .203 | .038 | .006 | .000 |
| 6                                              | 1.000 | 1.000 | 1.000 | 1.000 | .999  | .991 | .965 | .894 | .745 | .497 | .187 | .057 | .003 |
| 7                                              | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .999 | .996 | .983 | .942 | .832 | .570 | .337 | .077 |

e.  $n = 9$ 

| $\begin{smallmatrix} p \\ k \end{smallmatrix}$ | .01   | .05   | .10   | .20   | .30   | .40   | .50  | .60  | .70  | .80  | .90  | .95  | .99  |
|------------------------------------------------|-------|-------|-------|-------|-------|-------|------|------|------|------|------|------|------|
| 0                                              | .914  | .630  | .387  | .134  | .040  | .010  | .002 | .000 | .000 | .000 | .000 | .000 | .000 |
| 1                                              | .997  | .929  | .775  | .436  | .196  | .071  | .020 | .004 | .000 | .000 | .000 | .000 | .000 |
| 2                                              | 1.000 | .992  | .947  | .738  | .463  | .232  | .090 | .025 | .004 | .000 | .000 | .000 | .000 |
| 3                                              | 1.000 | .999  | .992  | .914  | .730  | .483  | .254 | .099 | .025 | .003 | .000 | .000 | .000 |
| 4                                              | 1.000 | 1.000 | .999  | .980  | .901  | .733  | .500 | .267 | .099 | .020 | .001 | .000 | .000 |
| 5                                              | 1.000 | 1.000 | 1.000 | .997  | .975  | .901  | .746 | .517 | .270 | .086 | .008 | .001 | .000 |
| 6                                              | 1.000 | 1.000 | 1.000 | 1.000 | .996  | .975  | .910 | .768 | .537 | .262 | .053 | .008 | .000 |
| 7                                              | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .996  | .980 | .929 | .804 | .564 | .225 | .071 | .003 |
| 8                                              | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .998 | .990 | .960 | .866 | .613 | .370 | .086 |

f.  $n = 10$ 

| $\begin{smallmatrix} p \\ k \end{smallmatrix}$ | .01   | .05   | .10   | .20   | .30   | .40   | .50  | .60  | .70  | .80  | .90  | .95  | .99  |
|------------------------------------------------|-------|-------|-------|-------|-------|-------|------|------|------|------|------|------|------|
| 0                                              | .904  | .599  | .349  | .107  | .028  | .006  | .001 | .000 | .000 | .000 | .000 | .000 | .000 |
| 1                                              | .996  | .914  | .736  | .376  | .149  | .046  | .011 | .002 | .000 | .000 | .000 | .000 | .000 |
| 2                                              | 1.000 | .988  | .930  | .678  | .383  | .167  | .055 | .012 | .002 | .000 | .000 | .000 | .000 |
| 3                                              | 1.000 | .999  | .987  | .879  | .650  | .382  | .172 | .055 | .011 | .001 | .000 | .000 | .000 |
| 4                                              | 1.000 | 1.000 | .998  | .967  | .850  | .633  | .377 | .166 | .047 | .006 | .000 | .000 | .000 |
| 5                                              | 1.000 | 1.000 | 1.000 | .994  | .953  | .834  | .623 | .367 | .150 | .033 | .002 | .000 | .000 |
| 6                                              | 1.000 | 1.000 | 1.000 | .999  | .989  | .945  | .828 | .618 | .350 | .121 | .013 | .001 | .000 |
| 7                                              | 1.000 | 1.000 | 1.000 | 1.000 | .998  | .988  | .945 | .833 | .617 | .322 | .070 | .012 | .000 |
| 8                                              | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .998  | .989 | .954 | .851 | .624 | .264 | .086 | .004 |
| 9                                              | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .999 | .994 | .972 | .893 | .651 | .401 | .096 |

(continued)

TABLE II (continued)

g.  $n = 15$ 

| $\begin{smallmatrix} p \\ k \end{smallmatrix}$ | .01   | .05   | .10   | .20   | .30   | .40   | .50   | .60   | .70  | .80  | .90  | .95  | .99  |
|------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|------|------|------|
| 0                                              | .860  | .463  | .206  | .035  | .005  | .000  | .000  | .000  | .000 | .000 | .000 | .000 | .000 |
| 1                                              | .990  | .829  | .549  | .167  | .035  | .005  | .000  | .000  | .000 | .000 | .000 | .000 | .000 |
| 2                                              | 1.000 | .964  | .816  | .398  | .127  | .027  | .004  | .000  | .000 | .000 | .000 | .000 | .000 |
| 3                                              | 1.000 | .995  | .944  | .648  | .297  | .091  | .018  | .002  | .000 | .000 | .000 | .000 | .000 |
| 4                                              | 1.000 | .999  | .987  | .838  | .515  | .217  | .059  | .009  | .001 | .000 | .000 | .000 | .000 |
| 5                                              | 1.000 | 1.000 | .998  | .939  | .722  | .403  | .151  | .034  | .004 | .000 | .000 | .000 | .000 |
| 6                                              | 1.000 | 1.000 | 1.000 | .982  | .869  | .610  | .304  | .095  | .015 | .001 | .000 | .000 | .000 |
| 7                                              | 1.000 | 1.000 | 1.000 | .996  | .950  | .787  | .500  | .213  | .050 | .004 | .000 | .000 | .000 |
| 8                                              | 1.000 | 1.000 | 1.000 | .999  | .985  | .905  | .696  | .390  | .131 | .018 | .000 | .000 | .000 |
| 9                                              | 1.000 | 1.000 | 1.000 | 1.000 | .996  | .966  | .849  | .597  | .278 | .061 | .002 | .000 | .000 |
| 10                                             | 1.000 | 1.000 | 1.000 | 1.000 | .999  | .991  | .941  | .783  | .485 | .164 | .013 | .001 | .000 |
| 11                                             | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .998  | .982  | .909  | .703 | .352 | .056 | .005 | .000 |
| 12                                             | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .996  | .973  | .873 | .602 | .184 | .036 | .000 |
| 13                                             | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .995  | .965 | .833 | .451 | .171 | .010 |
| 14                                             | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .995 | .965 | .794 | .537 | .140 |

h.  $n = 20$ 

| $\begin{smallmatrix} p \\ k \end{smallmatrix}$ | .01   | .05   | .10   | .20   | .30   | .40   | .50   | .60   | .70  | .80  | .90  | .95  | .99  |
|------------------------------------------------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|------|------|------|
| 0                                              | .818  | .358  | .122  | .012  | .001  | .000  | .000  | .000  | .000 | .000 | .000 | .000 | .000 |
| 1                                              | .983  | .736  | .392  | .069  | .008  | .001  | .000  | .000  | .000 | .000 | .000 | .000 | .000 |
| 2                                              | .999  | .925  | .677  | .206  | .035  | .004  | .000  | .000  | .000 | .000 | .000 | .000 | .000 |
| 3                                              | 1.000 | .984  | .867  | .411  | .107  | .016  | .001  | .000  | .000 | .000 | .000 | .000 | .000 |
| 4                                              | 1.000 | .997  | .957  | .630  | .238  | .051  | .006  | .000  | .000 | .000 | .000 | .000 | .000 |
| 5                                              | 1.000 | 1.000 | .989  | .804  | .416  | .126  | .021  | .002  | .000 | .000 | .000 | .000 | .000 |
| 6                                              | 1.000 | 1.000 | .998  | .913  | .608  | .250  | .058  | .006  | .000 | .000 | .000 | .000 | .000 |
| 7                                              | 1.000 | 1.000 | 1.000 | .968  | .772  | .416  | .132  | .021  | .001 | .000 | .000 | .000 | .000 |
| 8                                              | 1.000 | 1.000 | 1.000 | .990  | .887  | .596  | .252  | .057  | .005 | .000 | .000 | .000 | .000 |
| 9                                              | 1.000 | 1.000 | 1.000 | .997  | .952  | .755  | .412  | .128  | .017 | .001 | .000 | .000 | .000 |
| 10                                             | 1.000 | 1.000 | 1.000 | .999  | .983  | .872  | .588  | .245  | .048 | .003 | .000 | .000 | .000 |
| 11                                             | 1.000 | 1.000 | 1.000 | 1.000 | .995  | .943  | .748  | .404  | .113 | .010 | .000 | .000 | .000 |
| 12                                             | 1.000 | 1.000 | 1.000 | 1.000 | .999  | .979  | .868  | .584  | .228 | .032 | .000 | .000 | .000 |
| 13                                             | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .994  | .942  | .750  | .392 | .087 | .002 | .000 | .000 |
| 14                                             | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .998  | .979  | .874  | .584 | .196 | .011 | .000 | .000 |
| 15                                             | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .994  | .949  | .762 | .370 | .043 | .003 | .000 |
| 16                                             | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .999  | .984  | .893 | .589 | .133 | .016 | .000 |
| 17                                             | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .996  | .965 | .794 | .323 | .075 | .001 |
| 18                                             | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .999  | .992 | .931 | .608 | .264 | .017 |
| 19                                             | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | .999 | .988 | .878 | .642 | .182 |

(continued)

**TABLE XIV** Critical Values of  $T_L$  and  $T_U$  for the Wilcoxon Rank Sum Test: Independent Samples

*Test statistic is the rank sum associated with the smaller sample (if equal sample sizes, either rank sum can be used).*

**a.  $\alpha = .025$  one-tailed;  $\alpha = .05$  two-tailed**

| $n_2 \backslash n_1$ | 3     |       | 4     |       | 5     |       | 6     |       | 7     |       | 8     |       | 9     |       | 10    |       |
|----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                      | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ |
| <b>3</b>             | 5     | 16    | 6     | 18    | 6     | 21    | 7     | 23    | 7     | 26    | 8     | 28    | 8     | 31    | 9     | 33    |
| <b>4</b>             | 6     | 18    | 11    | 25    | 12    | 28    | 12    | 32    | 13    | 35    | 14    | 38    | 15    | 41    | 16    | 44    |
| <b>5</b>             | 6     | 21    | 12    | 28    | 18    | 37    | 19    | 41    | 20    | 45    | 21    | 49    | 22    | 53    | 24    | 56    |
| <b>6</b>             | 7     | 23    | 12    | 32    | 19    | 41    | 26    | 52    | 28    | 56    | 29    | 61    | 31    | 65    | 32    | 70    |
| <b>7</b>             | 7     | 26    | 13    | 35    | 20    | 45    | 28    | 56    | 37    | 68    | 39    | 73    | 41    | 78    | 43    | 83    |
| <b>8</b>             | 8     | 28    | 14    | 38    | 21    | 49    | 29    | 61    | 39    | 73    | 49    | 87    | 51    | 93    | 54    | 98    |
| <b>9</b>             | 8     | 31    | 15    | 41    | 22    | 53    | 31    | 65    | 41    | 78    | 51    | 93    | 63    | 108   | 66    | 114   |
| <b>10</b>            | 9     | 33    | 16    | 44    | 24    | 56    | 32    | 70    | 43    | 83    | 54    | 98    | 66    | 114   | 79    | 131   |

**b.  $\alpha = .05$  one-tailed;  $\alpha = .10$  two-tailed**

| $n_2 \backslash n_1$ | 3     |       | 4     |       | 5     |       | 6     |       | 7     |       | 8     |       | 9     |       | 10    |       |
|----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                      | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ | $T_L$ | $T_U$ |
| <b>3</b>             | 6     | 15    | 7     | 17    | 7     | 20    | 8     | 22    | 9     | 24    | 9     | 27    | 10    | 29    | 11    | 31    |
| <b>4</b>             | 7     | 17    | 12    | 24    | 13    | 27    | 14    | 30    | 15    | 33    | 16    | 36    | 17    | 39    | 18    | 42    |
| <b>5</b>             | 7     | 20    | 13    | 27    | 19    | 36    | 20    | 40    | 22    | 43    | 24    | 46    | 25    | 50    | 26    | 54    |
| <b>6</b>             | 8     | 22    | 14    | 30    | 20    | 40    | 28    | 50    | 30    | 54    | 32    | 58    | 33    | 63    | 35    | 67    |
| <b>7</b>             | 9     | 24    | 15    | 33    | 22    | 43    | 30    | 54    | 39    | 66    | 41    | 71    | 43    | 76    | 46    | 80    |
| <b>8</b>             | 9     | 27    | 16    | 36    | 24    | 46    | 32    | 58    | 41    | 71    | 52    | 84    | 54    | 90    | 57    | 95    |
| <b>9</b>             | 10    | 29    | 17    | 39    | 25    | 50    | 33    | 63    | 43    | 76    | 54    | 90    | 66    | 105   | 69    | 111   |
| <b>10</b>            | 11    | 31    | 18    | 42    | 26    | 54    | 35    | 67    | 46    | 80    | 57    | 95    | 69    | 111   | 83    | 127   |

Source: From F. Wilcoxon, and R. A. Wilcox. "Some Rapid Approximate Statistical Procedures," 1964, 20–23. Courtesy of Lederle Laboratories Division of American Cyanamid Company, Madison, NJ.